

# First Normal Form (1NF) Relational Database Design

> **Definition:** A relation is in **First Normal Form (1NF)** if and only if all attributes contain only **atomic (indivisible) values**, and there are no repeating groups or arrays stored within a single field.

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## 1. Detailed Technical Explanation

Normalization is the systematic process of organizing relational database schemas to eliminate data redundancy and anomalies (Insertion, Deletion, and Update anomalies).

### Key Rules of 1NF:

1. **Atomic Values:** Each column must store a single value (no multi-valued attributes like comma-separated phone numbers).

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### Conversion Example: Unnormalized Table to 1NF

#### Unnormalized Relation (UNF):

| Stud

#### Normalized 1NF Relation:

| Stud

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## 2. Core Concepts & Memory Keywords

- **Atomic Domains:** Fields storing single indivisible values.

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## 3. Must-Write Points for Exams

- 1NF eliminates **multi-valued attributes** and **composite attributes** from relation instances.

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#### 4. Quick Recall Flow

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